

Attached is a paper that was published in June 2016 titled "The Effectiveness of Class Size Reduction" by Dr. William J. Mathis. It is an independent academic review of the research that has been done on class size and the relationship to student performance and outcomes. Currently Dr. Mathis is Vice-Chairman of the Vermont State Board of Education.

Some excerpts from his publication:

*Ask a parent if they want their child in a class of 15 or a class of 25. The answer is predictable. Intuitively, they know that smaller classes will provide more personalized attention, a better climate, and result in more learning. Ask teachers, and they will wax eloquent on the importance of small classes in providing individual support to their students.*

*There are many studies of the impact of smaller classes and they vary widely in quality. As a result, proponents from all perspectives can cherry-pick studies that support their point of view. But let's look closer. There is, in fact, an independent consensus on what we know. The most prominent study supporting smaller class sizes was the Tennessee STAR (Student/Teacher Achievement Ratio) experiment. The STAR experiment was a four-year statewide random-assignment experiment. Students in Kindergarten in the same schools were randomly assigned to classes of 13-15; to classes of 22-25 with a teacher's aide; or to classes of 25 without a teacher's aide. In the early studies, these students were followed through grade 3. In practice, the small classes ranged in size from 13-18 and the large classes from 22-28. It is worth noting that even the larger classes were smaller than most classes in those grades in Tennessee at the time. The smaller classes performed substantially better by the end of second grade in test scores, grades, and fewer disciplinary referrals. The gains lasted. The students that had been assigned to smaller classes were more likely to graduate in four years, more likely to go to college, and more likely to get a degree in a STEM field. The positive effect was twice as large for poor and minority students, and thus narrowed the achievement gap.*

Dr. Mathis cites 28 references in his paper and has these conclusions and recommendations:

- *There is little evidence indicating that other reforms would be more effective at a lower cost.*
- *Class size is an important determinant of student outcomes, and one that can be directly determined by policy. All else being equal, lowering class sizes will improve student outcomes.*
- *The payoff from class-size reduction is greater for low-income and minority children. Conversely, increases in class size are likely to be especially harmful to these populations -- who are already more likely to be subjected to large classes.*